

Development-Aware Flood Susceptibility Mapping Using Multi-Temporal Satellite Data

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Introduction

- Floods are India's most frequent and deadliest natural disaster. Kerala alone lost 400+ lives since the 2018 floods, with 1.2 million people evacuated



Introduction

- **Flooding** is a major natural hazard that affects people, infrastructure, and the environment.
- **Urbanization** is rapidly changing land use, especially in cities and surrounding regions.
- **Satellite imagery** provides a way to observe floods and changes in land use over time.
- **Machine Learning** can be used to analyse this spatial data and identify areas vulnerable to flooding.
- **This project studies the relationship between urban growth and flood susceptibility in Kerala.**

Literature Review

Event-time flood detection & hydraulic modelling

Paper (Year)	Method	Key Result	Key Limitation
Mohamadiazar et al. (2024) – J. Hydrology [1]	CNN (U-Net) on Sentinel-1 SAR	AUC 0.93, F1 89.85%	Needs live rainfall; no per-factor explainability
Antzoulatos et al. (2022) – Sustainability [2]	Random Forest (best), SVM, Naive Bayes, NN	F1 $\approx$ 0.99	Needs calibrated 2D hydraulic model for labels
Flood Susceptibility Mapping (2024) – Zhu et al [12]	Sentinel-1 GRID SAR, XGBoost, SVC	Farmland accounted for 54.8% of the flooded area. EMF model accuracy: 0.942 training and 0.940 testing	Does not explicitly model how urban development changes flood susceptibility over time.
Al Mehedi et al. (2022) – Scientific Reports [10]	LR / DT / SVM / KNN classify; DNN regresses depth	DT F1 0.991; depth RMSE 0.027	Surrogate for a calibrated hydraulic model; needs live discharge

Literature Review

Static susceptibility mapping

Paper (Year)	Method	Key Result	Key Limitation
Waleed & Sajjad (2025) — IJDRR [3]	XGBoost/LightGBM + SHAP	AUC 0.93, F1 89.85%	Self-stated: does not integrate land use change as a driver, named explicitly as future work
Kendagannaswamy et al. (2025) — Discover Appl. Sci. [4]	LightGBM + AHP (dual model)	LightGBM AUC 0.948 vs AHP 0.769; 18.5% of Kerala very vulnerable	Self-stated: conventional random split, no spatial or temporal CV; SRTM canopy bias; single static snapshot
Islam et al. (2025) - A Systematic Review of Urban Flood Susceptibility Mapping [6]	Comparative survey: MCDA, FR, LR, SVM, RF, ANN, CNN	ML consistently beats expert weighted MCDA	Contributes no benchmark of its own; stops Sept 2024; does not treat validation design as a first-class issue

Literature Review

Scenario & change-driven susceptibility

Paper (Year)	Method	Key Result	Key Limitation
Khodaei et al. (2025) - Future flood susceptibility mapping under climate and land use change [7]	RF, XGBoost, SVC — each PSO-tuned; CA Markov + CMIP6	RF AUC 0.98; high-risk zone 20%;	Random negative sampling and a plain 70/30 split; LULC bundled with climate, so the pure development effect is never isolated
Land Cover Change Effects on Current and Future Flood Susceptibility (2024) - Rashidiyan et al. [13]	LULC, altitude, slope , Logistic Regression	SVM best: RMSE 0.125, R^2 0.9373	Future-LULC simulation adds a second modelling layer on top of the flood model itself
Impact of climate and land use change on flood vulnerability (2025) - Abijith et al. [9]	Sentinel-1, Random Forest; LULC projected to 2030 / 2040 / 2050	Moderate and very-high risk areas grow across every future projection	largely based on average annual precipitation and does not explicitly model the downstream effects of upstream development.

Literature Review

Methodology (labels, sampling and honest validation)

Paper (Year)	Method	Key Result	Key Limitation
Zhang et al. (2025) - Remote Sensing [5]	FR + One-Class SVM; RF, AdaBoost, GBDT, SHAP	AUC varied 0.71–0.95 depending on negative sampling	Only one rural basin, its own 0.95 still comes from a random split
Li et al. (2022) — A Positive-Unlabeled Learning Algorithm [11]	PBLC (PU Learning), benchmarked against ANN	AUC $\approx$ 0.90 for both, but PBLC far better on Fpb; calibrated AUC $\approx$ 0.96	Could not test PBLC with a CNN (only 532 positives); omits soil, geology and drainage density of data
Koldasbayeva et al. (2024) - Nature Communications [8]	Spatial validation & uncertainty analysis	Random splits can give over-optimistic results	General geospatial study, not flood-specific

Motivation

- Urban growth is accelerating flood risk faster than climate change alone, every acre of paved/roofed surface sheds water instead of absorbing it
- Kochi/Periyar basin: dense monsoon-driven urbanization, low-lying terrain, and rapid construction make it a high-stakes test case
- Flood susceptibility mapping (FSM) using satellite imagery + ML has become the standard disaster-preparedness tool worldwide
- But existing FSM tools answer "where has it flooded before" not "what happens to this land once it's built on"

Key Novelty & Contributions

Our Contribution	Key Novelty
Development-Aware Prediction	Predicts current vs. post-development flood susceptibility
UICA / Δ UICA	Captures upstream urbanization $\rightarrow$ downstream runoff
Natural-Experiment Validation	Uses observed 2018–2021 urban growth for validation
Leak-Free Evaluation	Uses spatial CV + temporal hold-out
Counterfactual Recommendations	Provides actionable ways to reduce vulnerability
Positive-Unlabeled Learning	Does not assume “not flooded” = “safe”

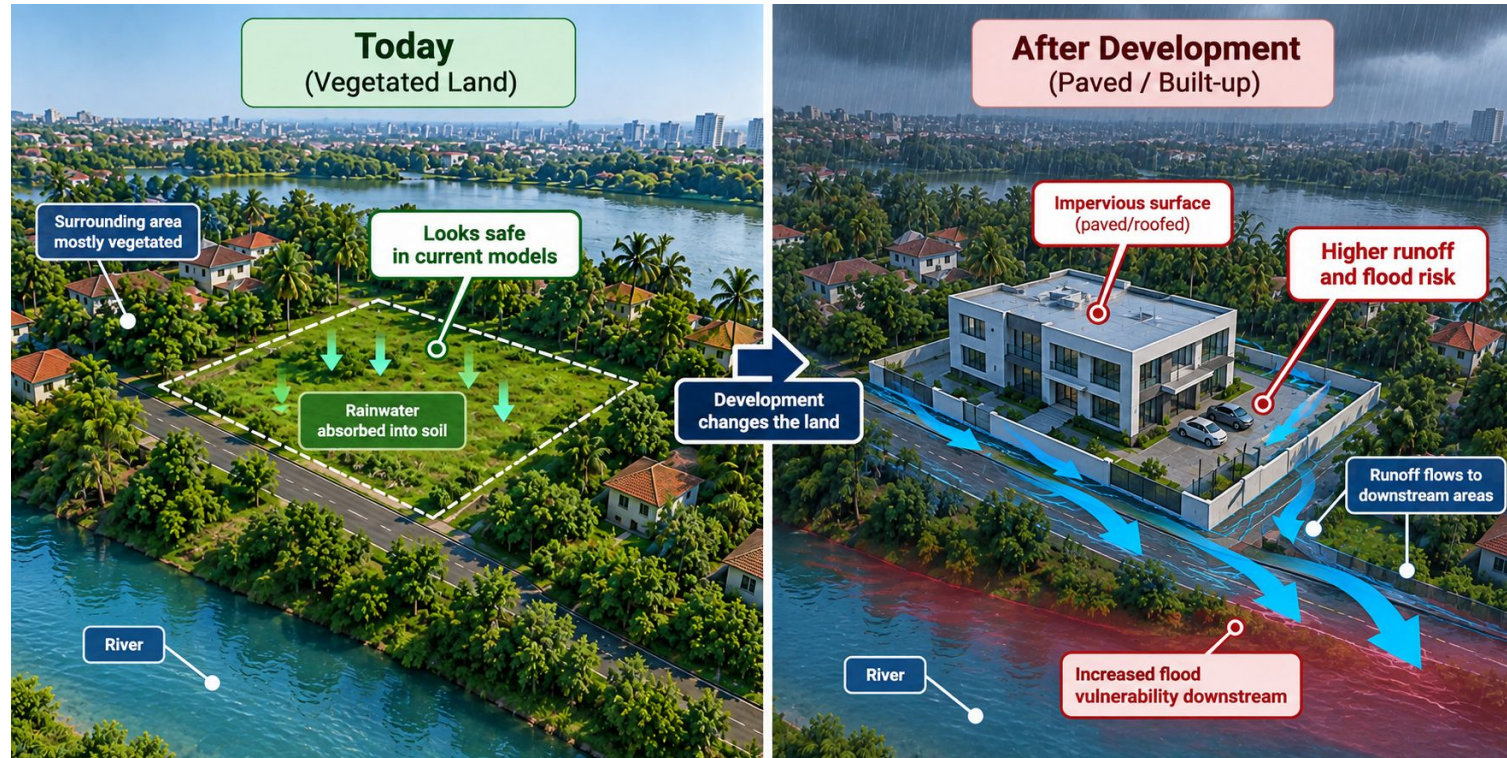
Problem Statement

Existing flood susceptibility models primarily assess risk based on the current state of the land, without considering how future urban development may alter flood vulnerability or how upstream development can influence downstream runoff.

They assess where flooding may occur based on existing land conditions, but do not adequately capture:

- **Future development** of currently undeveloped land
- **Upstream → downstream effects** of new impervious surfaces
- **Reliable spatial and temporal validation** of susceptibility predictions

Problem Statement

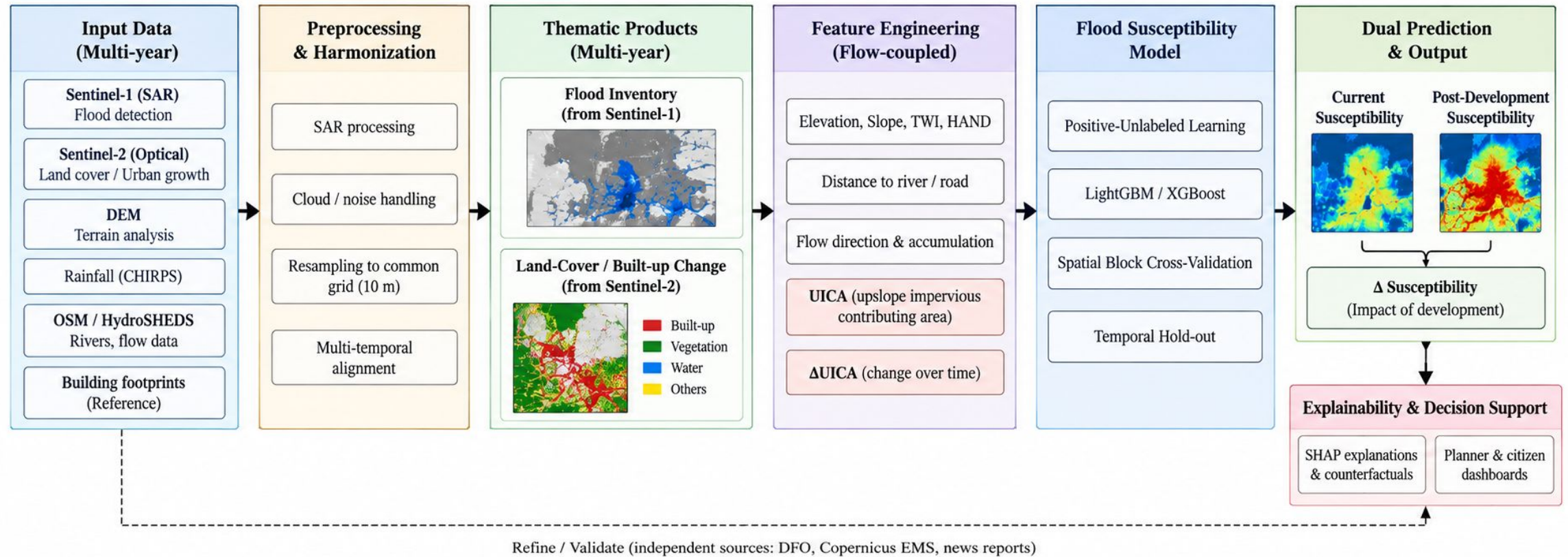


Therefore, there is a need for a development-aware approach that can estimate how flood susceptibility changes with urban growth.

Objectives

1. **Develop a multi-year flood inventory** using satellite data.
2. **Identify urban growth and land-cover changes** across multiple years.
3. **Capture upstream development effects** using flow-based features such as UICA and Δ UICA.
4. **Develop a flood susceptibility model** using machine learning with reliable spatial-temporal validation.
5. **Compare current and post-development susceptibility** to assess the impact of urban growth.
6. **Provide explainable and actionable outputs** to support flood-risk planning and pre-construction decisions.

System Architecture



Datasets

Data Type	Dataset	What it provides	Purpose in our Project
Optical	Sentinel-2 L2A	Multispectral bands	Land-cover & urban-growth detection
Radar	Sentinel-1 GRD (VV + VH)	Radar backscatter	Flood inventory, works through clouds
Elevation	FABDEM v1.2	Bare-earth elevation	Terrain & flow routing
Rainfall	CHIRPS Daily	Daily precipitation	Long-term rainfall characteristics
Water	JRC Global Surface Water	Permanent water occurrence	Remove permanent water from flood mapping

Data Type	Dataset	What it provides	Purpose in our Project
Hydrology	HydroBASINS L12	Sub-watershed boundaries	Spatial CV blocks
Features	HydroRIVERS + OSM	Rivers, roads & infrastructure	Distance-based features
Buildings	Microsoft / Google Open Buildings / OSM	Building footprints	Built-up labels & change detection
Flood Evidence	DFO + Copernicus EMS/UNOSAT	Observed flood extents	Independent validation

Evaluation

1. Flood Mapping / Segmentation

Used to evaluate the quality of predicted flood extent:

- **IoU (Intersection over Union)** — overlap between predicted and actual flood areas
- **Dice / F1 Score** — similarity between predicted and reference flood masks
- **Precision** — proportion of predicted flooded pixels that are actually flooded
- **Recall** — proportion of actual flooded pixels correctly detected

3. Model Comparison & Validation

We will compare:

- **Baseline vs. Proposed Model**
- **Random Split vs. Spatial Block Cross-Validation**
- **Conventional vs. PU Learning**
- **Current vs. Post-Development Susceptibility**

2. Flood Susceptibility Model

Used to evaluate the ML model:

- **ROC-AUC** — ability to distinguish high- and low-susceptibility locations
- **PR-AUC** — useful when flooded locations are much fewer than non-flooded/unlabelled locations
- **F1 Score** — balance between precision and recall
- **Precision & Recall** — detailed classification performance

Conclusion

Base paper finalized: Mohamadiazar et al. (2024)

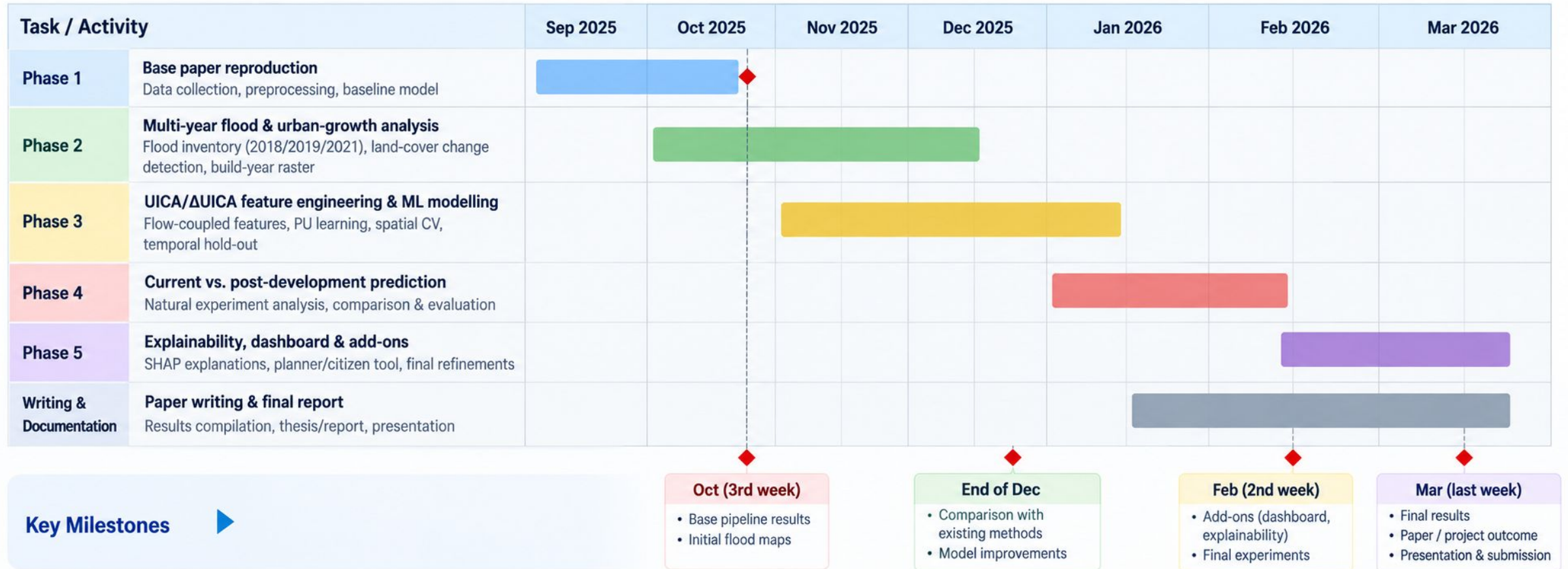
Research gap identified: Need for **development-aware flood susceptibility assessment**

Proposed approach: Multi-temporal data + **flow-coupled features** + ML

Key outcome: Compare **current vs. post-development** flood susceptibility

Goal: Support **explainable and informed urban planning decisions**

Project Timeline & Planned Work



References

- [1] N. Mohamadiazar, A. Ebrahimian, and H. Hosseiny, "Integrating deep learning, satellite image processing, and spatial-temporal analysis for urban flood prediction," *Journal of Hydrology*, vol. 639, p. 131508, 2024.
- [2] G. Antzoulatos, I.-O. Kouloglou, M. Bakratsas, A. Moumtzidou, I. Gialampoukidis, A. Karakostas, F. Lombardo, R. Fiorin, D. Norbiato, M. Ferri, A. Symeonidis, S. Vrochidis, and I. Kompatsiaris, "Flood hazard and risk mapping by applying an explainable machine learning framework using satellite imagery and GIS data," *Sustainability*, vol. 14, no. 6, p. 3251, 2022.
- [3] M. Waleed and M. Sajjad, "High-resolution flood susceptibility mapping and exposure assessment in Pakistan: An integrated artificial intelligence, machine learning and geospatial framework," *International Journal of Disaster Risk Reduction*, vol. 121, p. 105442, 2025.
- [4] M. S. Kendagannaswamy, C. K. Roopa, B. S. Harish, and M. S. Mukesh, "Multi-criteria decision analysis for regional-scale flood susceptibility mapping in Kerala state, India," *Discover Applied Sciences*, vol. 7, p. 598, 2025.
- [5] Y. Zhang, Y. Wei, R. Yao, P. Sun, N. Zhen, and X. Xia, "Data uncertainty of flood susceptibility using non-flood samples," *Remote Sens.*, vol. 17, no. 3, Art. no. 375, Jan. 2025, doi: 10.3390/rs17030375.
- [6] T. Islam, E. B. Zeleke, M. Afroz, and A. M. Melesse, "A systematic review of urban flood susceptibility mapping: Remote sensing, machine learning, and other modeling approaches," *Remote Sens.*, vol. 17, no. 3, Art. no. 524, Feb. 2025, doi: 10.3390/rs17030524.

References

- [7] H. Khodaei, F. Nasiri Saleh, A. Nobakht Dalir, and E. Zarei, "Future flood susceptibility mapping under climate and land use change," *Sci. Rep.*, vol. 15, Art. no. 12394, Apr. 2025, doi: 10.1038/s41598-025-97008-0.
- [8] D. Koldasbayeva, P. Tregubova, M. Gasanov, A. Zaytsev, A. Petrovskaya, and E. Burnaev, "Challenges in data-driven geospatial modeling for environmental research and practice," *Nat. Commun.*, vol. 15, Art. no. 10700, Dec. 2024, doi: 10.1038/s41467-024-55240-8.
- [9] D. Abijith, S. Saravanan, K. S. S. Parthasarathy, N. M. Reddy, J. Niraimathi, A. A. Bindajam, J. Mallick, M. M. Alharbi, and H. G. Abdo, "Assessing the impact of climate and land use change on flood vulnerability: a machine learning approach in coastal region of Tamil Nadu, India," *Geoscience Letters*, vol. 12, p. 1, 2025.
- [10] M. A. Al Mehedi, V. Smith, H. Hosseiny, and X. Jiao, "Unraveling the complexities of urban fluvial flood hydraulics through AI," *Scientific Reports*, vol. 12, p. 18738, 2022.
- [11] W. Li, Y. Liu, Z. Liu, Z. Gao, H. Huang, and W. Huang, "A positive-unlabeled learning algorithm for urban flood susceptibility modeling," *Land*, vol. 11, no. 11, p. 1971, 2022.
- [12] X. Zhu, H. Guo, and J. J. Huang, "Urban flood susceptibility mapping using remote sensing, social sensing and an ensemble machine learning model," *Sustainable Cities and Society*, vol. 108, p. 105508, 2024.
- [13] M. Rashidiyan and M. Rahimzadegan, "Investigation and Evaluation of Land Use–Land Cover Change Effects on Current and Future Flood Susceptibility," *Natural Hazards Review*, vol. 25, no. 1, p. 04023049, 2024.